

ERA5 Wave and Wind Statistical Report

Input file: input.csv

Variables: swh, mwp, mwd, wind, dwi, u10, v10

1) Descriptive statistics

Index	count	mean	std	min	1%	5%	25%	50%	75%	95%	99%	max	skewness	kurtosis
swh	756960	2.213	1.058	0.390	0.710	0.930	1.450	1.990	2.740	4.260	5.650	11.080	1.328	2.644
mwp	756960	8.733	2.034	4	5.200	5.800	7.100	8.500	10.100	12.400	13.800	17.400	0.453	-0.356
mwd	756960	302.449	47.814	0	10	237	289	310	330	349	356	359	-3.572	18.437
wind	756960	6.926	3.368	0.010	0.830	1.860	4.290	6.690	9.350	12.690	14.920	28.950	0.325	-0.441
dwi	756960	204.321	127.286	0	0	4	77	225	331	355	359	359	-0.364	-1.345
u10	756960	1.071	3.770	-19.110	-7.850	-4.840	-1.150	0.920	2.930	7.990	11.660	24.340	0.383	1.088
v10	756960	-2.429	6.168	-23.710	-13.380	-11.430	-7.190	-3.060	1.740	8.770	12.490	27.950	0.407	-0.419

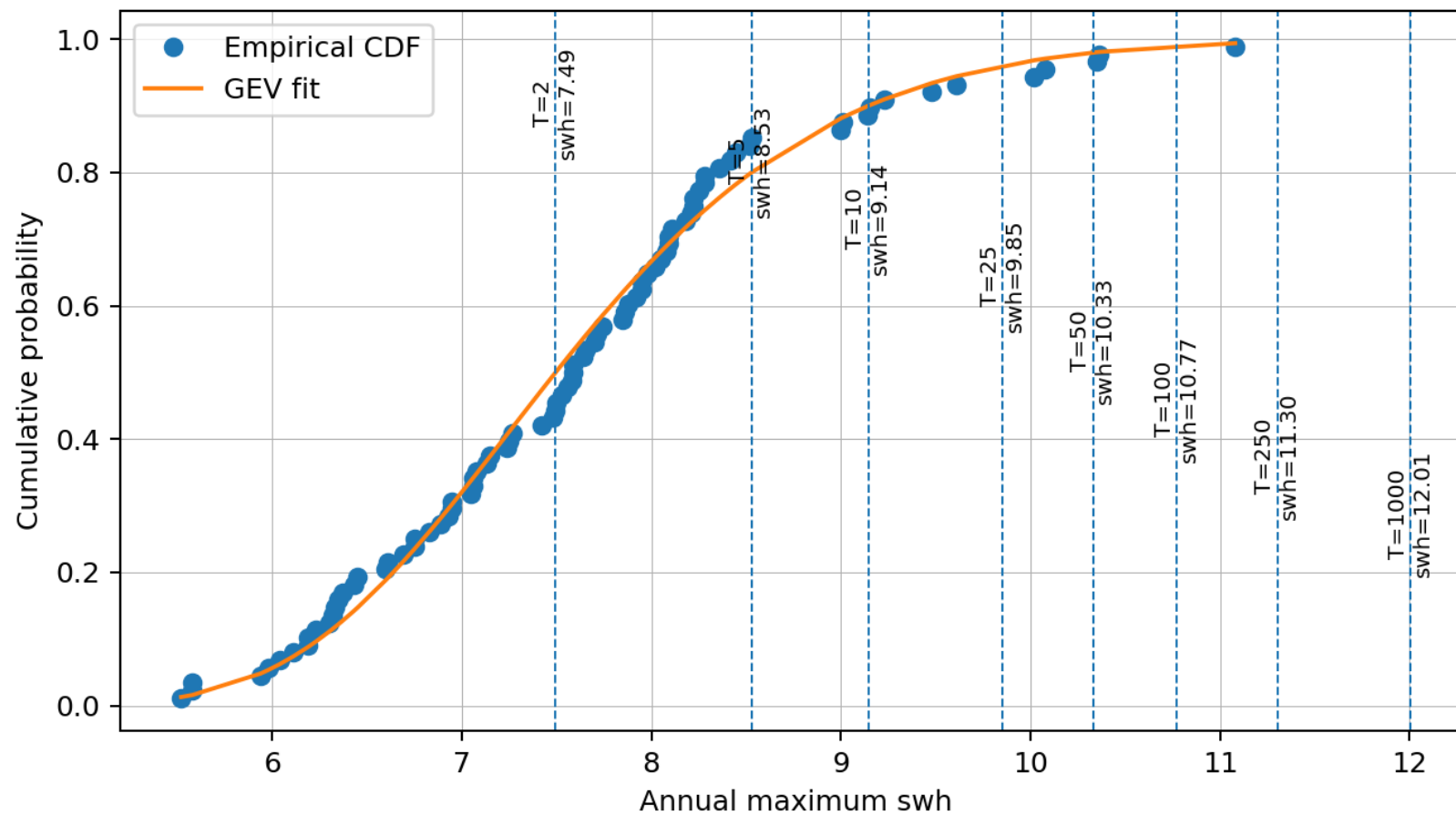
2) GEV: swl, all directions

Annual maxima used: 87 | shape=0.1111, loc=7.1320, scale=1.0106

Return levels

Index	Return level swl
2	7.49
5	8.53
10	9.14
25	9.85
50	10.33
100	10.77
250	11.30
1000	12.01

GEV fit: SWH - all directions



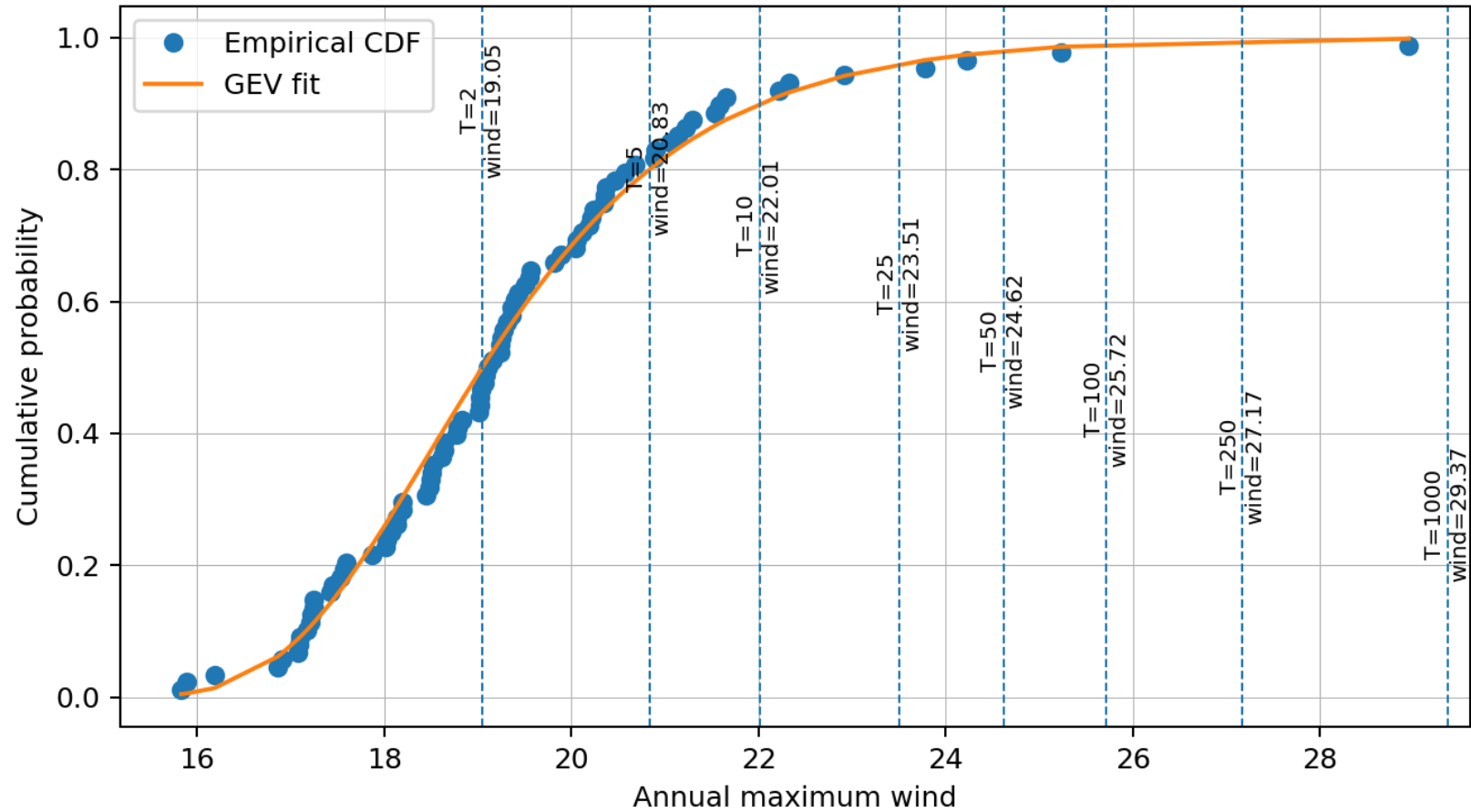
3) GEV: wind, all directions

Annual maxima used: 87 | shape=-0.0009, loc=18.4726, scale=1.5720

Return levels

Index	Return level wind
2	19.05
5	20.83
10	22.01
25	23.51
50	24.62
100	25.72
250	27.17
1000	29.37

GEV fit: WIND - all directions

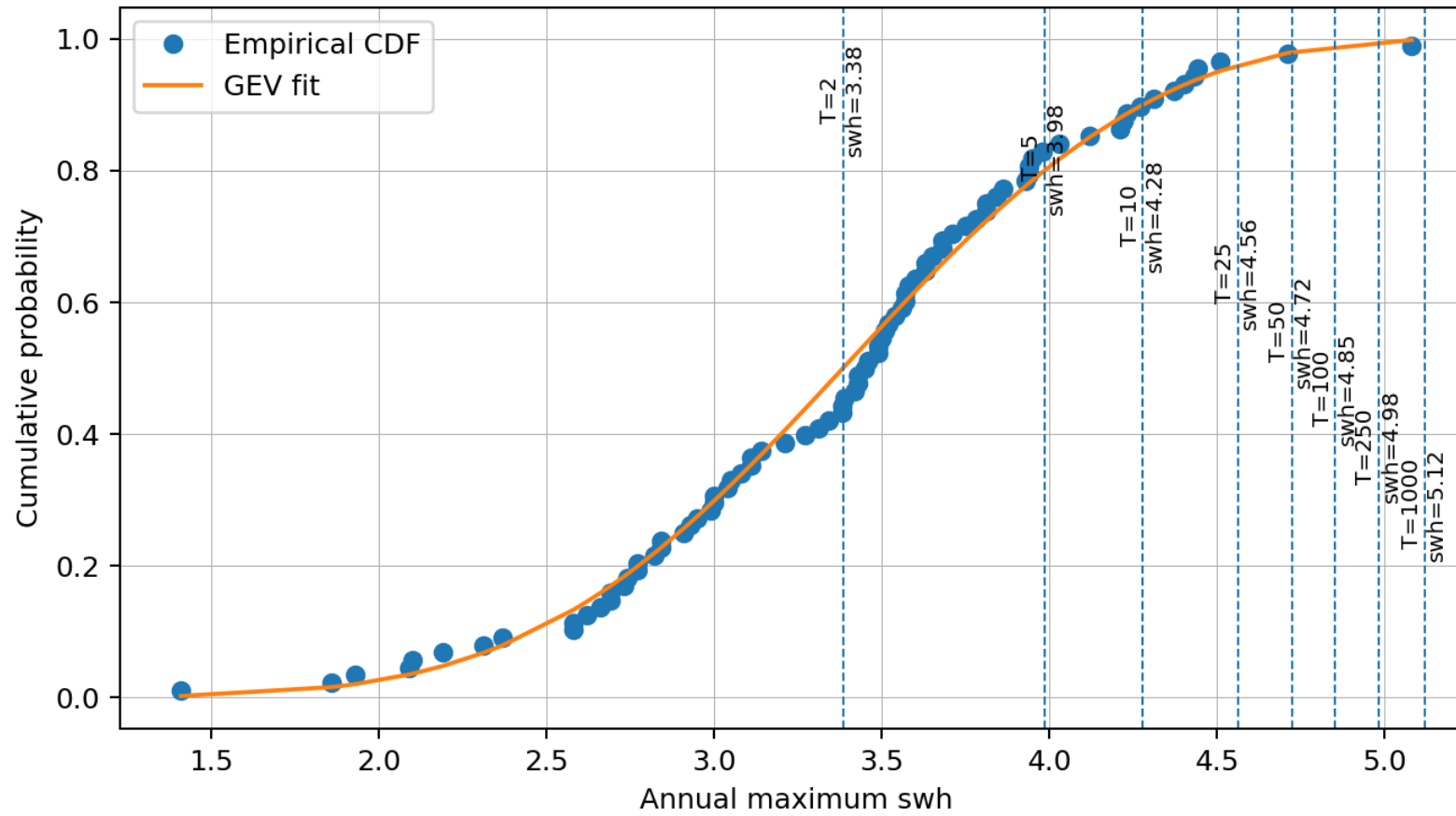


4) GEV: swl by 30-degree wave-direction sector

Index	n_annual_maxima	shape	loc	scale	RP_2	RP_5	RP_10	RP_25	RP_50	RP_100	RP_250	RP_1000
000-030	87	0.319	3.138	0.710	3.384	3.985	4.278	4.562	4.723	4.851	4.982	5.119
030-060	64	-0.037	1.774	0.359	1.907	2.329	2.617	2.994	3.282	3.575	3.973	4.599
060-090	49	-0.036	1.707	0.444	1.871	2.392	2.749	3.214	3.570	3.932	4.423	5.195
090-120	47	-0.028	1.765	0.508	1.952	2.543	2.944	3.464	3.858	4.257	4.794	5.631
120-150	61	0.128	1.950	0.626	2.174	2.804	3.173	3.592	3.872	4.125	4.427	4.819
150-180	74	0.124	2.914	1.030	3.283	4.324	4.937	5.633	6.100	6.524	7.030	7.691
180-210	87	-0.004	3.927	0.845	4.237	5.198	5.837	6.648	7.251	7.851	8.645	9.848
210-240	87	-0.052	4.764	0.845	5.077	6.083	6.783	7.708	8.424	9.162	10.176	11.801
240-270	87	0.065	5.729	1.127	6.137	7.340	8.089	8.986	9.616	10.214	10.961	12.009
270-300	87	0.242	6.720	1.200	7.141	8.230	8.803	9.394	9.752	10.052	10.378	10.750
300-330	87	0.129	6.445	1.012	6.807	7.825	8.421	9.095	9.545	9.953	10.437	11.066
330-360	87	0.073	4.975	0.714	5.233	5.990	6.457	7.013	7.400	7.766	8.220	8.850

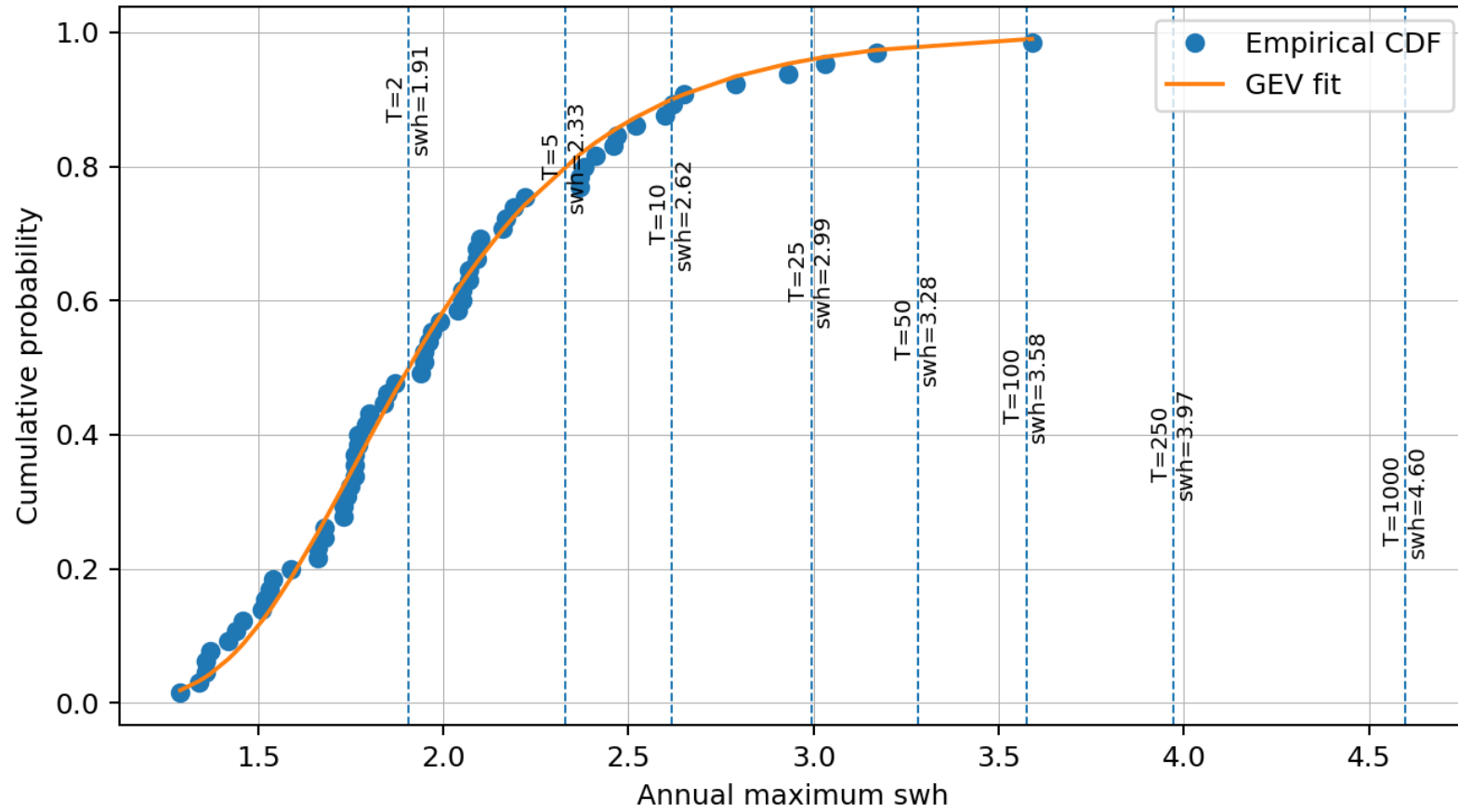
SWH sector 000-030

GEV fit: SWH - 000-030



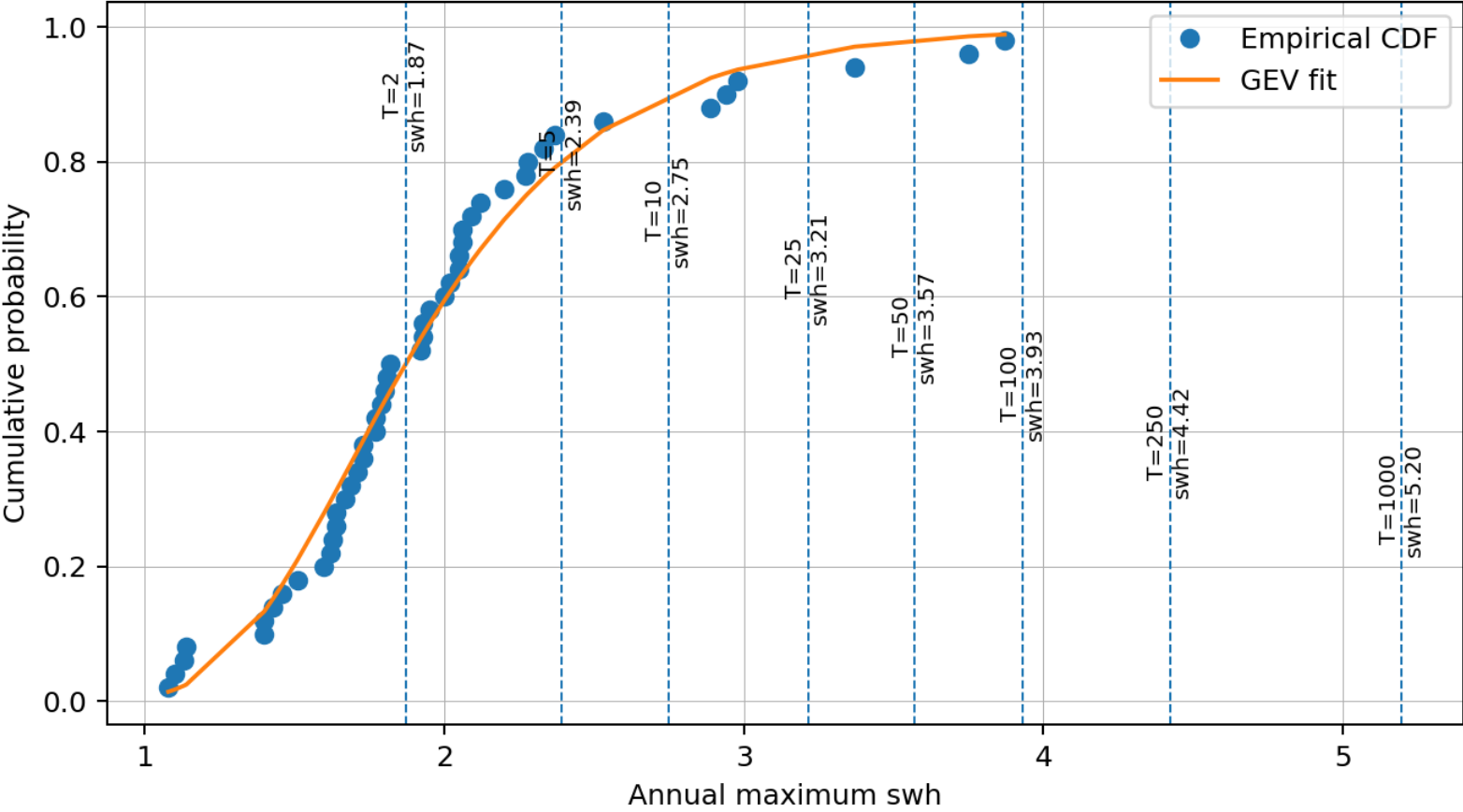
SWH sector 030-060

GEV fit: SWH - 030-060



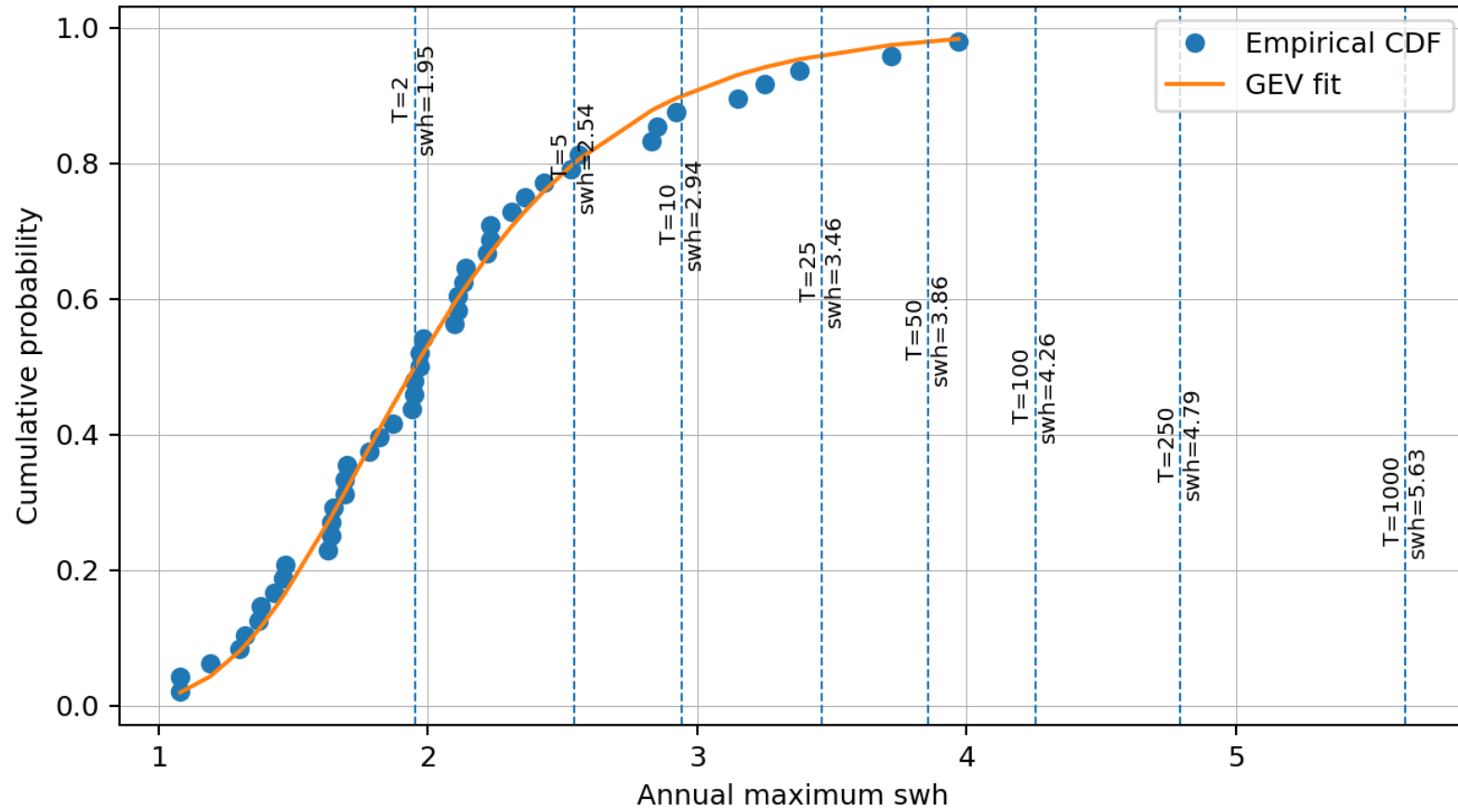
SWH sector 060-090

GEV fit: SWH - 060-090

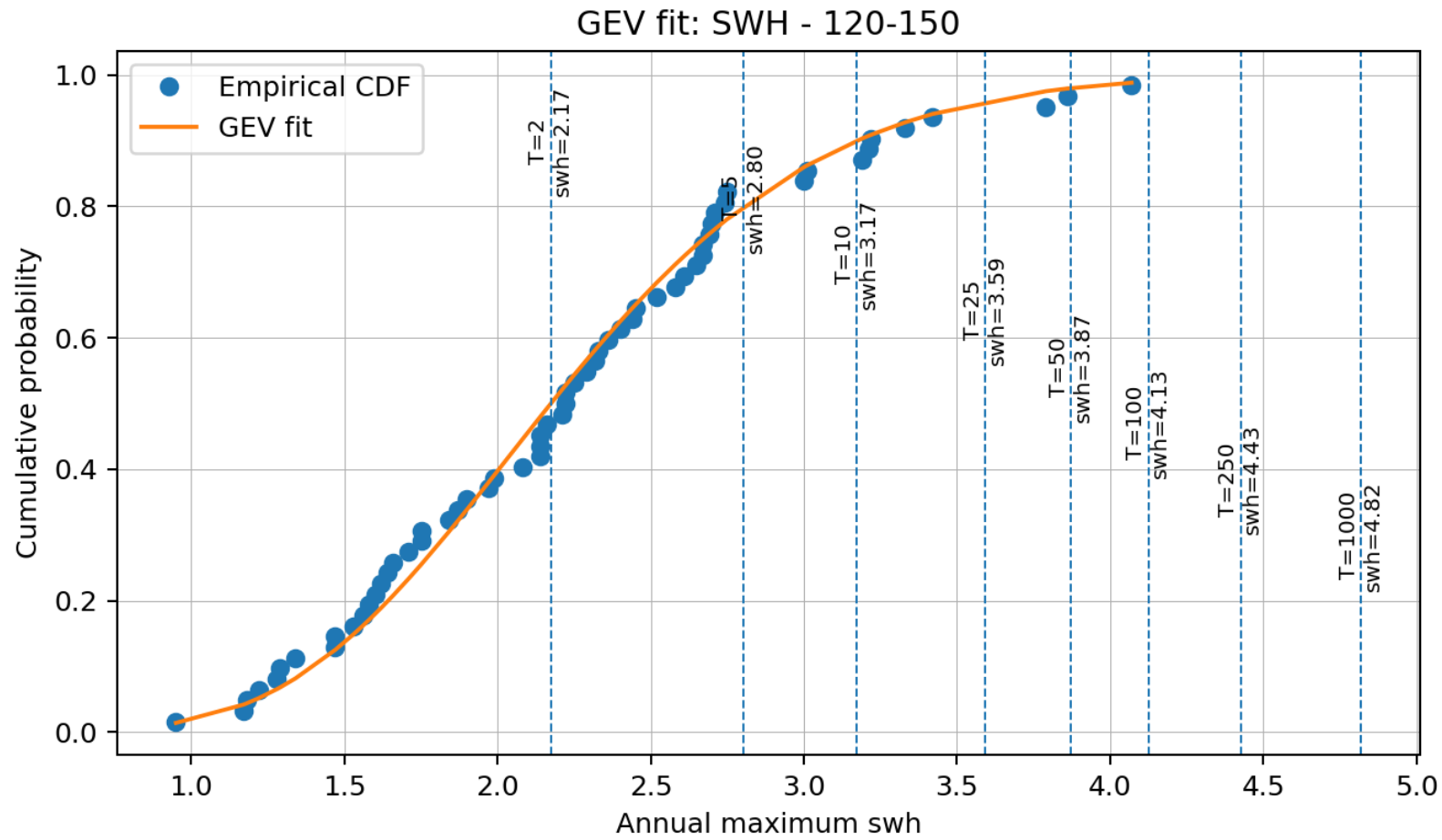


SWH sector 090-120

GEV fit: SWH - 090-120

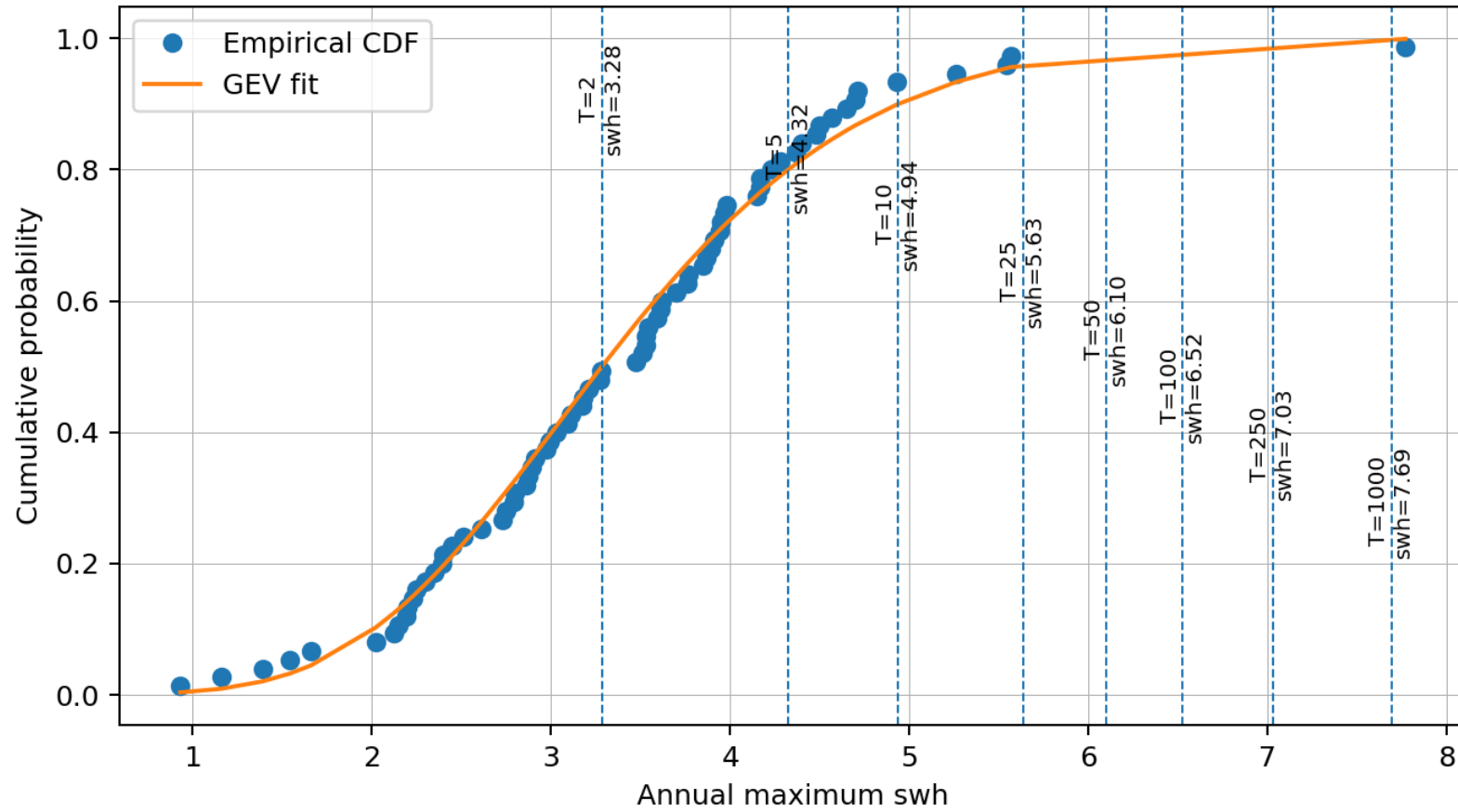


SWH sector 120-150

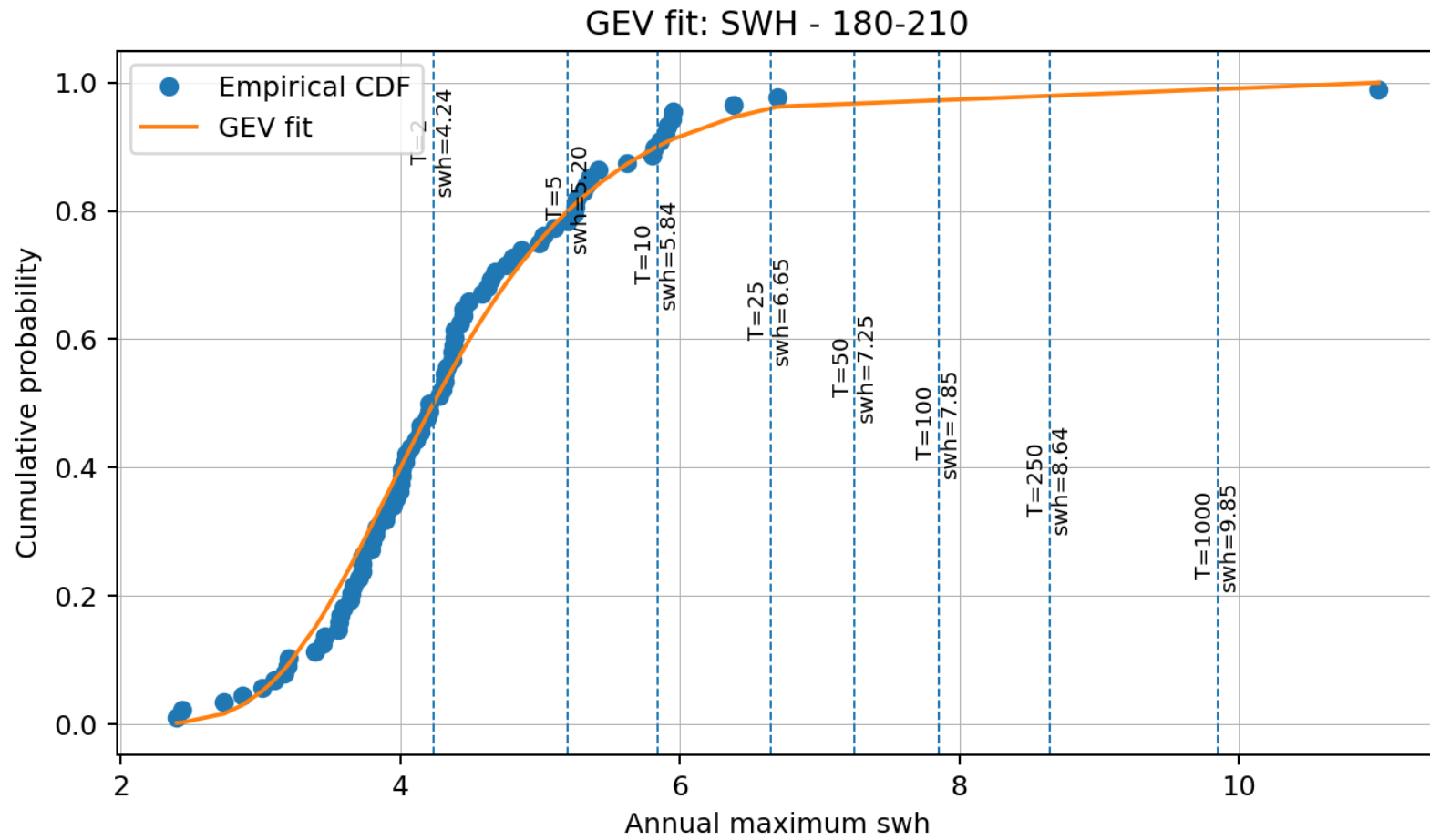


SWH sector 150-180

GEV fit: SWH - 150-180

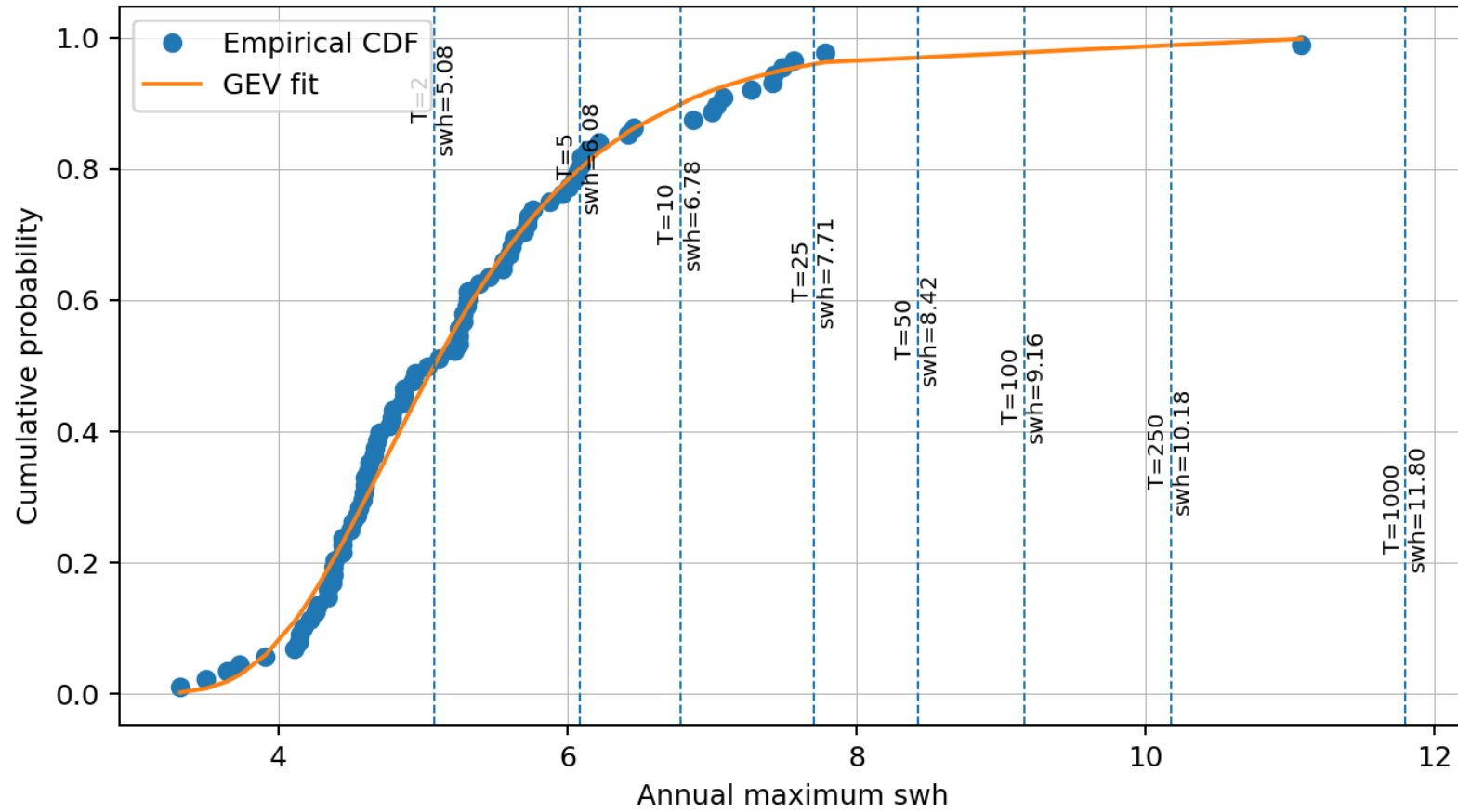


SWH sector 180-210



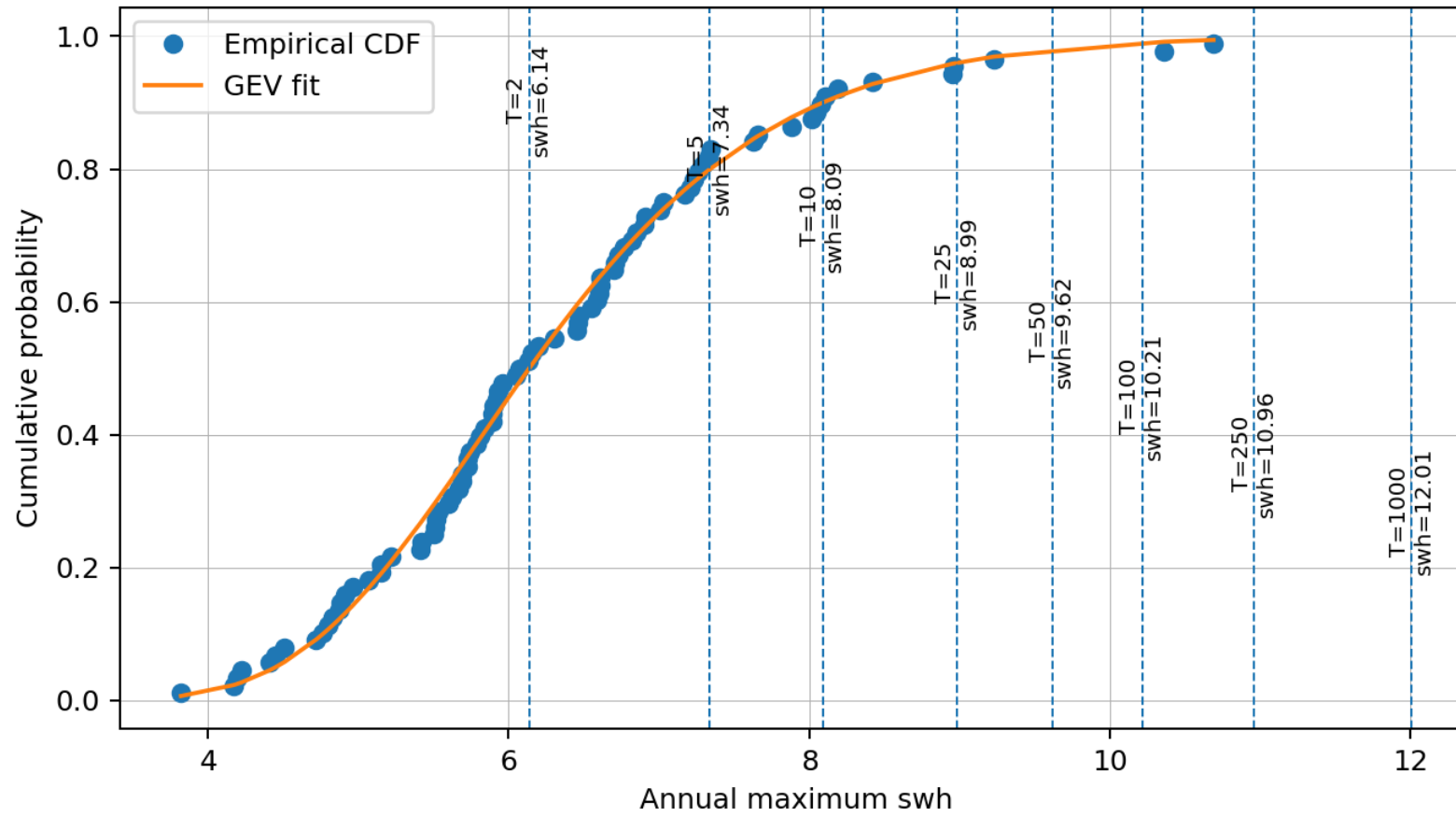
SWH sector 210-240

GEV fit: SWH - 210-240



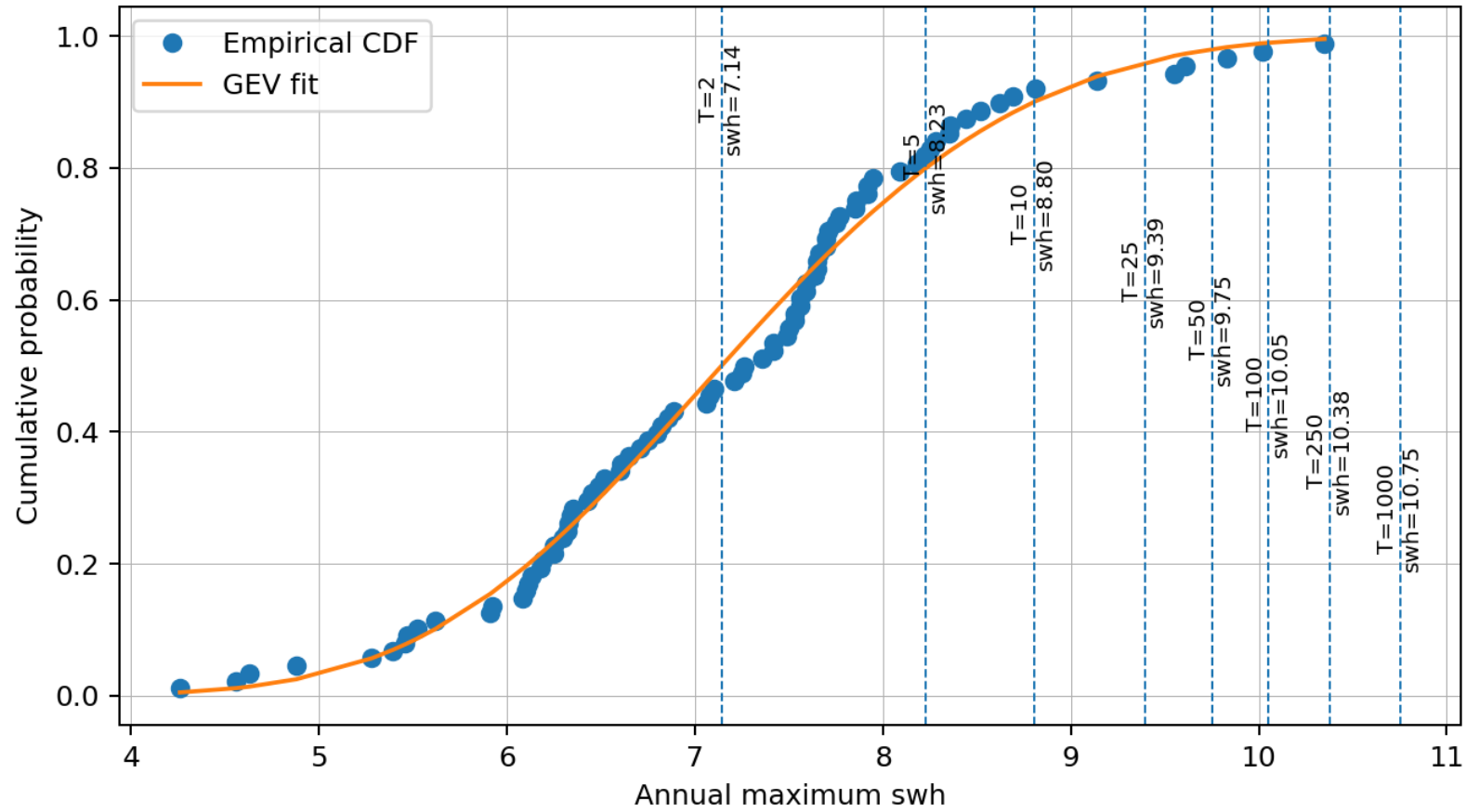
SWH sector 240-270

GEV fit: SWH - 240-270



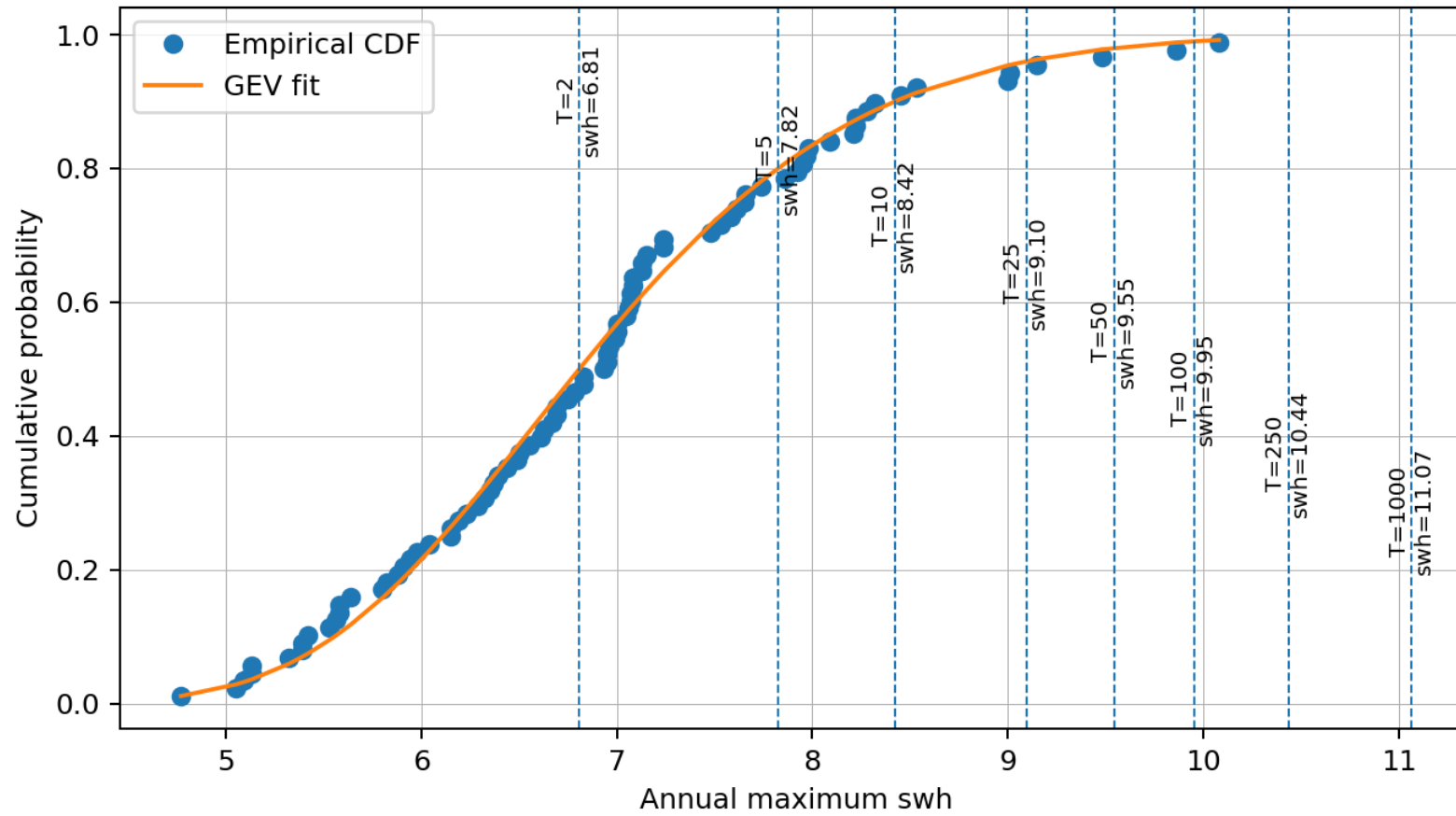
SWH sector 270-300

GEV fit: SWH - 270-300



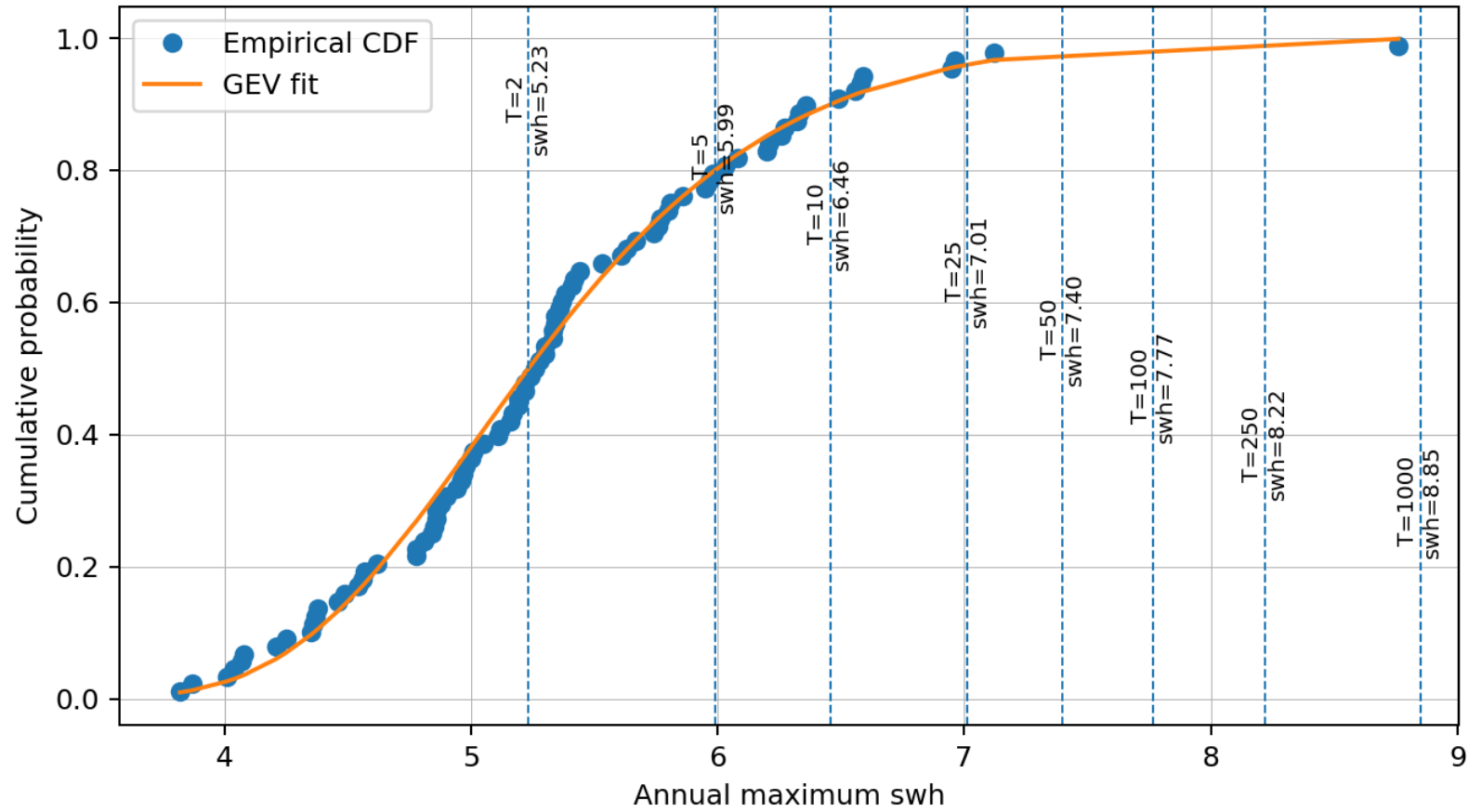
SWH sector 300-330

GEV fit: SWH - 300-330



SWH sector 330-360

GEV fit: SWH - 330-360

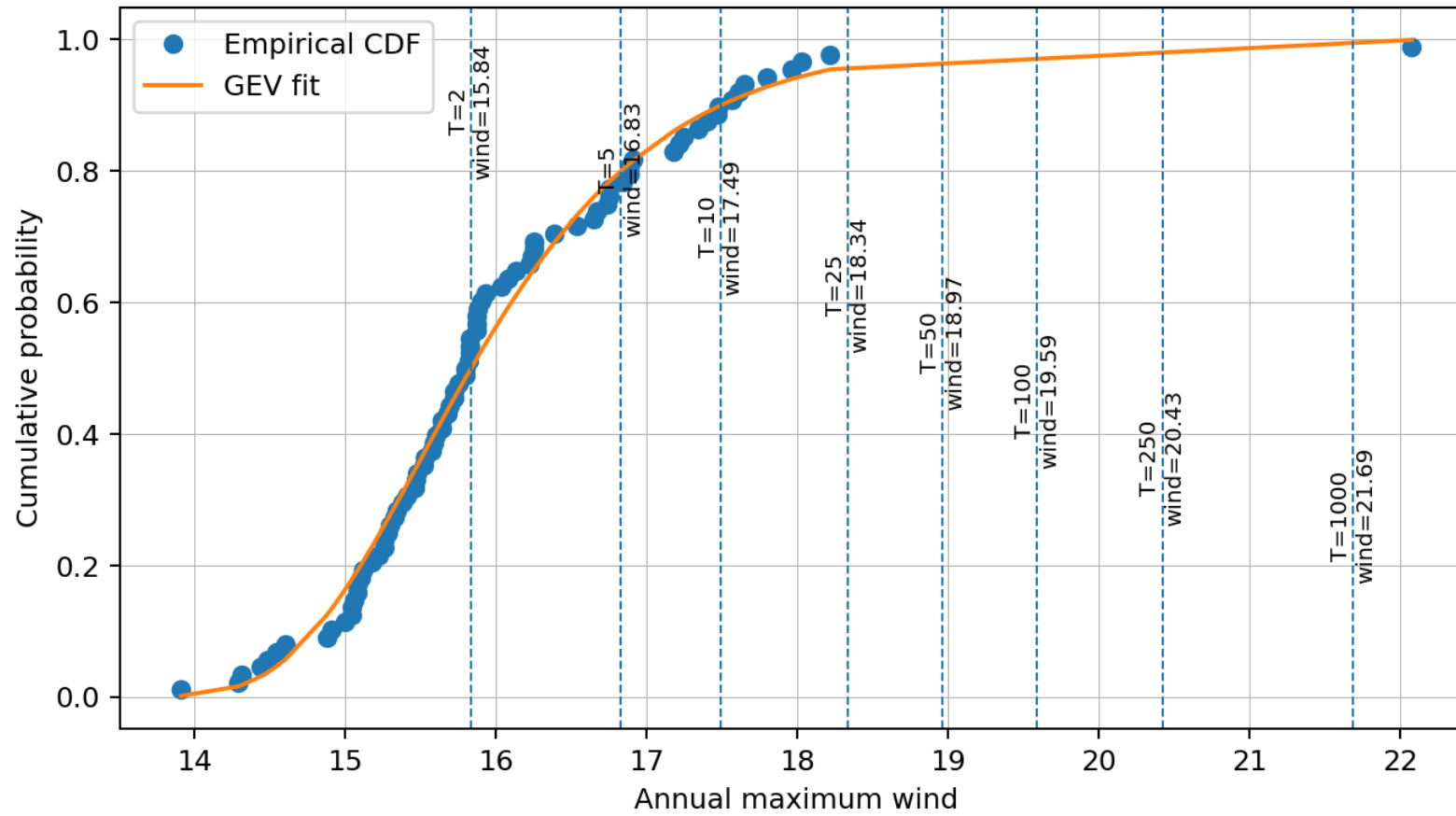


5) GEV: wind by 30-degree wind-direction sector

Index	n_annual_maxima	shape	loc	scale	RP_2	RP_5	RP_10	RP_25	RP_50	RP_100	RP_250	RP_1000
000-030	87	-0.007	15.516	0.871	15.836	16.830	17.493	18.337	18.966	19.594	20.426	21.693
030-060	87	0.264	12.254	1.453	12.762	14.054	14.720	15.393	15.795	16.126	16.478	16.872
060-090	87	0.222	10.448	1.274	10.896	12.073	12.703	13.364	13.772	14.118	14.499	14.946
090-120	87	-0.055	10.753	1.262	11.220	12.726	13.777	15.168	16.248	17.363	18.899	21.370
120-150	87	2.029	21.504	3.380	22.378	23.091	23.153	23.167	23.169	23.170	23.170	23.170
150-180	87	0.035	16.504	1.776	17.151	19.099	20.346	21.877	22.979	24.047	25.414	27.396
180-210	87	0.090	16.729	1.708	17.345	19.125	20.208	21.475	22.348	23.162	24.158	25.513
210-240	87	0.056	16.615	1.552	17.177	18.847	19.895	21.159	22.054	22.908	23.982	25.504
240-270	87	0.137	16.544	1.815	17.193	19.005	20.058	21.243	22.027	22.734	23.567	24.641
270-300	87	0.244	16.324	1.896	16.988	18.706	19.608	20.536	21.099	21.569	22.079	22.661
300-330	87	0.112	15.763	2.116	16.523	18.684	19.972	21.451	22.452	23.370	24.474	25.940
330-360	87	-0.137	15.693	1.255	16.165	17.783	19.002	20.733	22.170	23.741	26.052	30.142

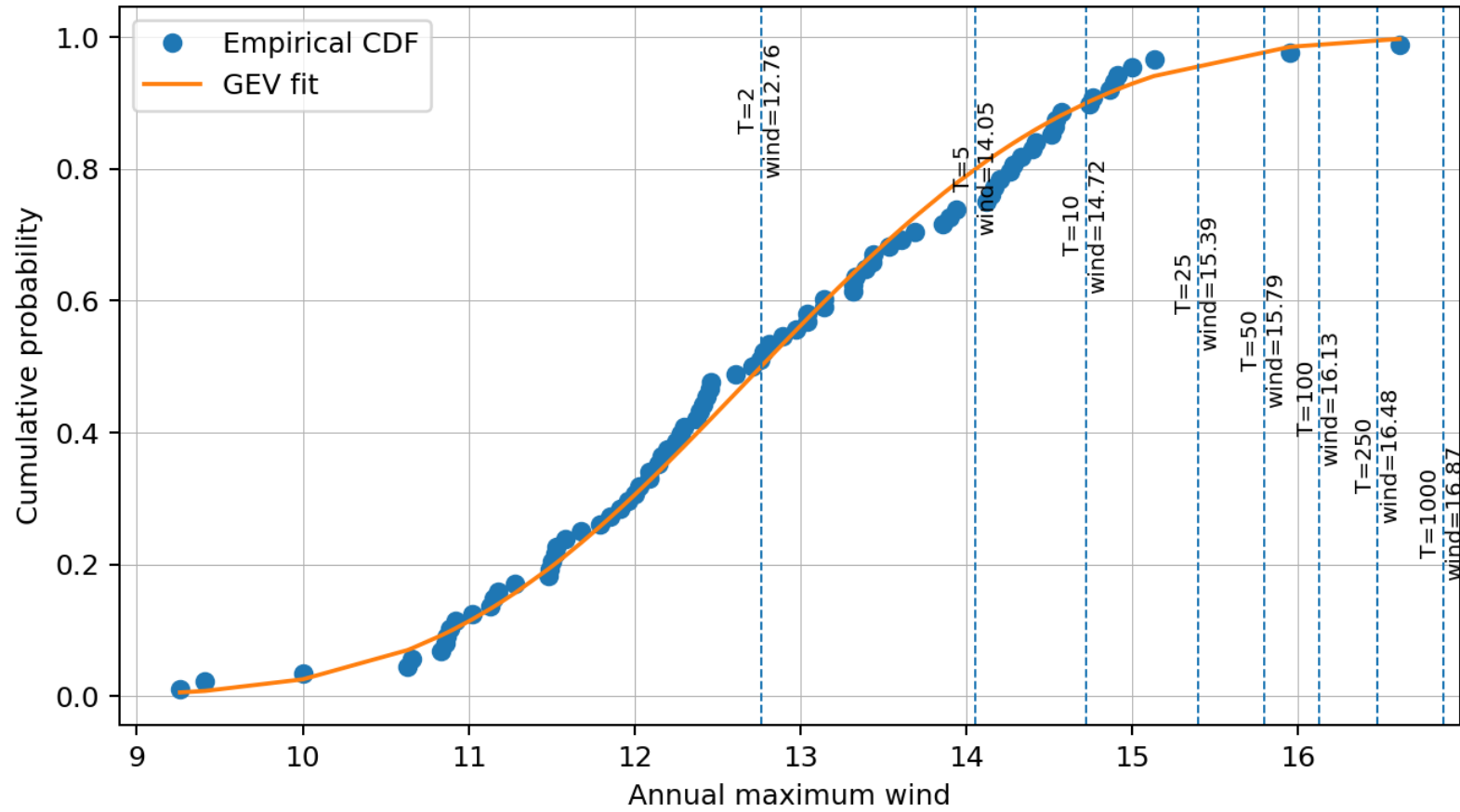
Wind sector 000-030

GEV fit: WIND - 000-030

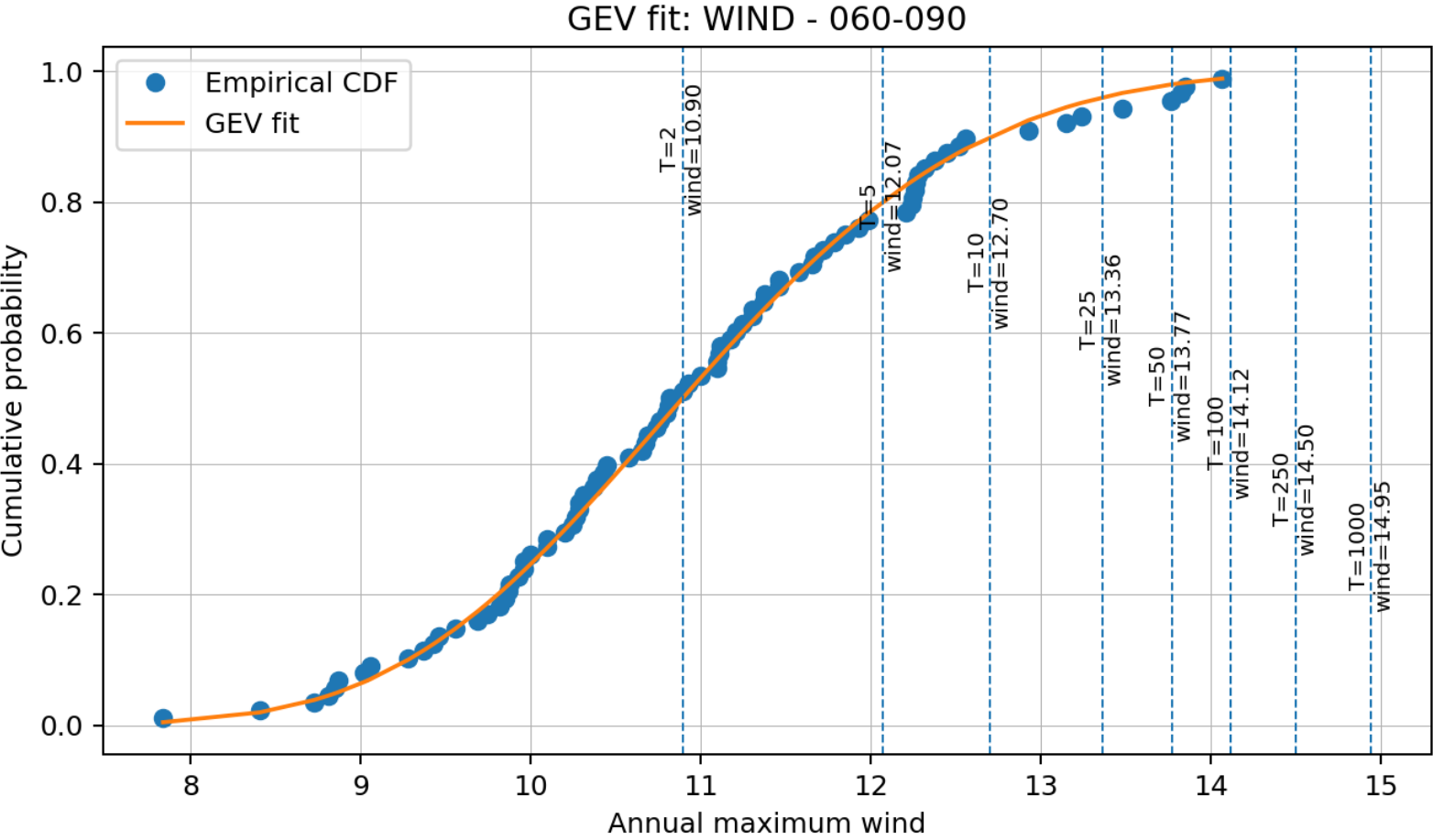


Wind sector 030-060

GEV fit: WIND - 030-060

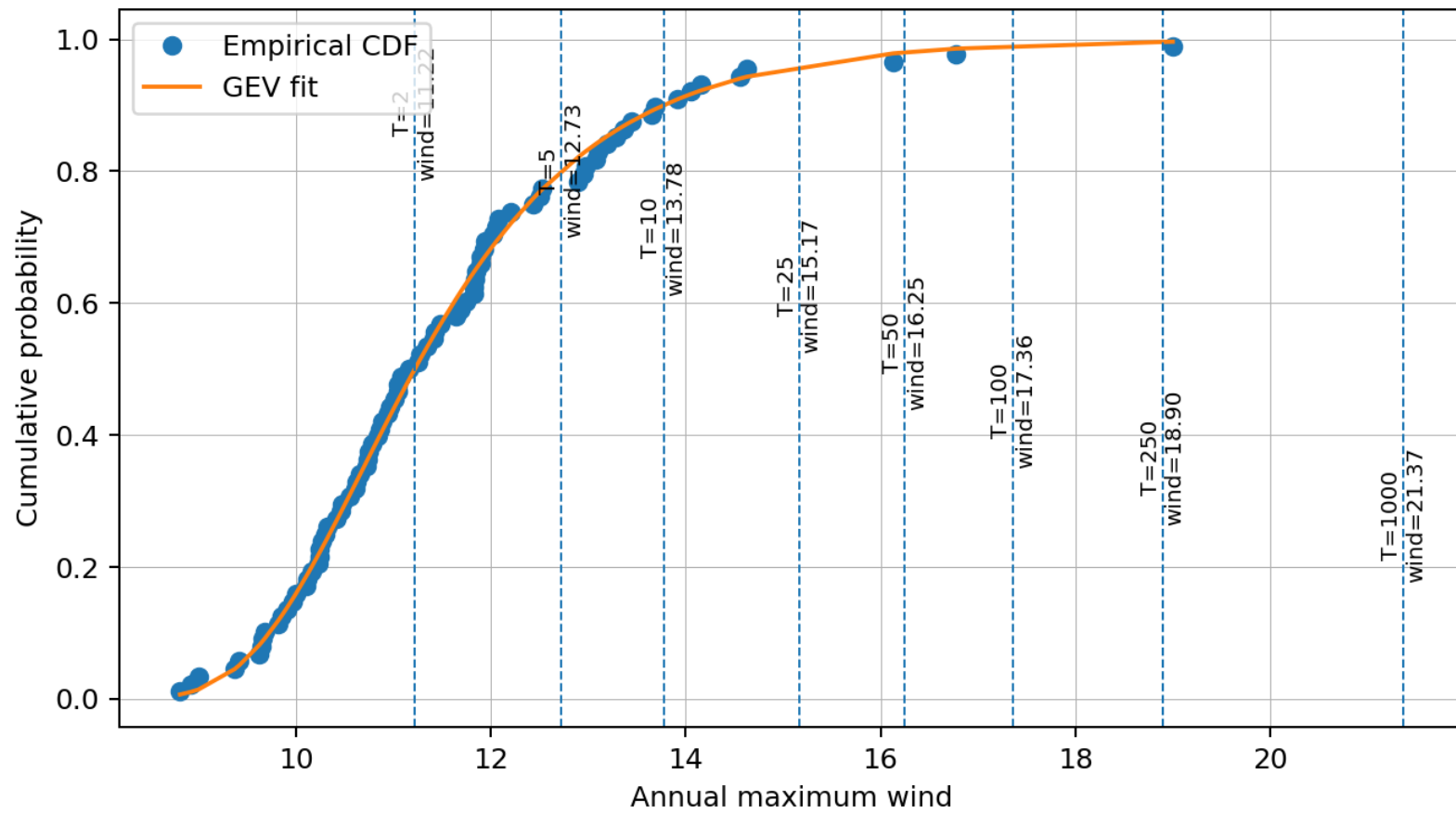


Wind sector 060-090

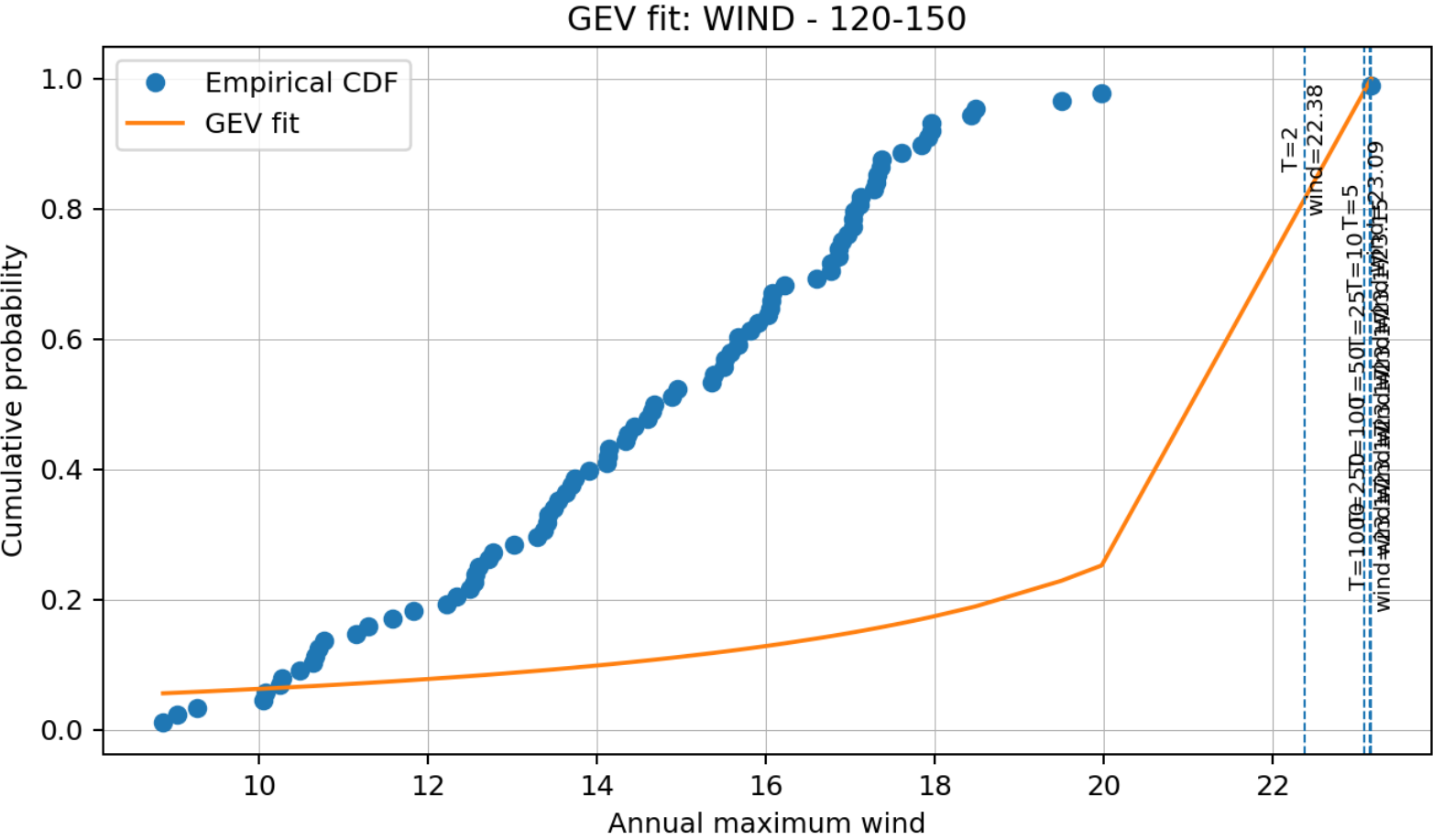


Wind sector 090-120

GEV fit: WIND - 090-120

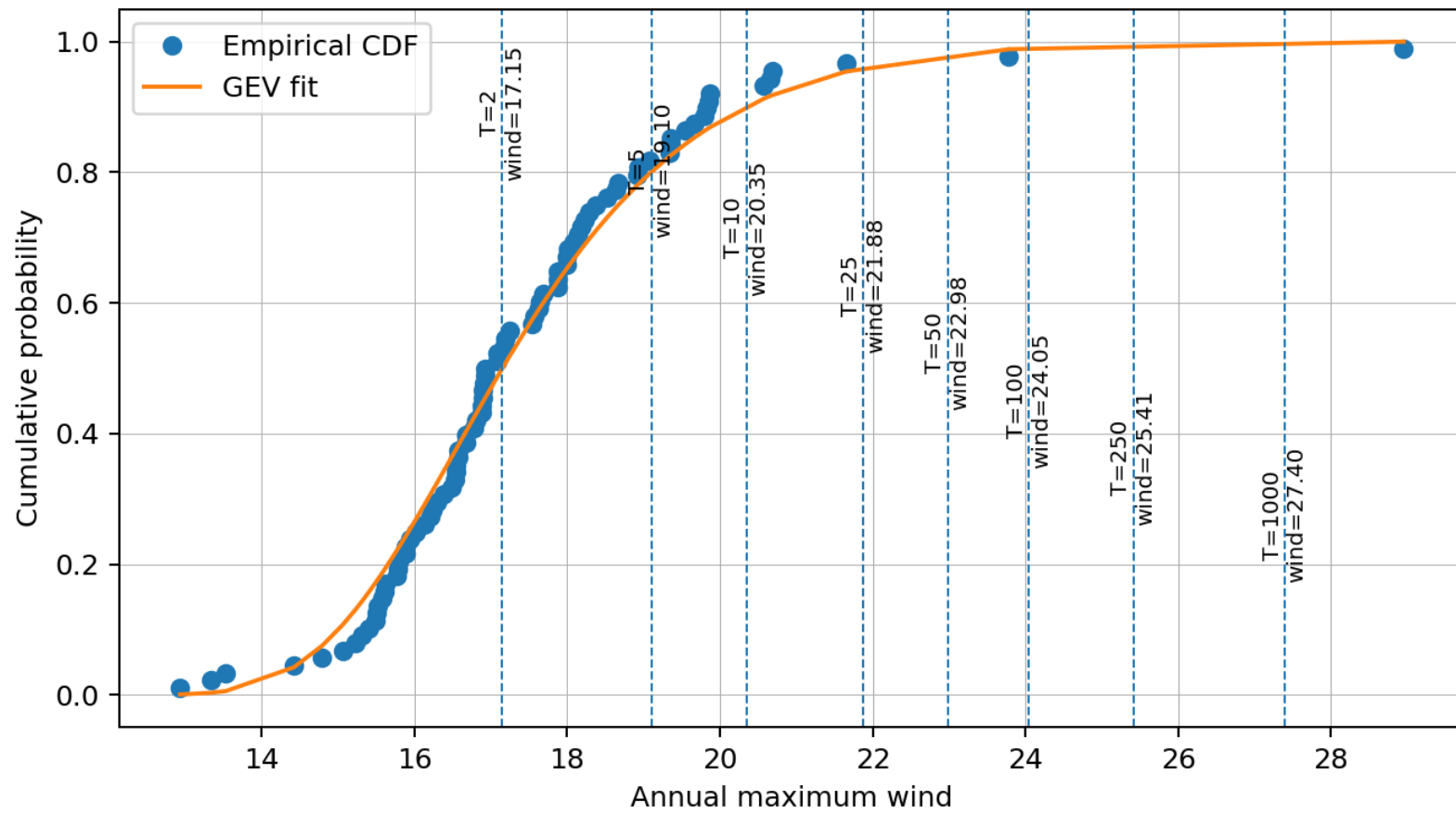


Wind sector 120-150

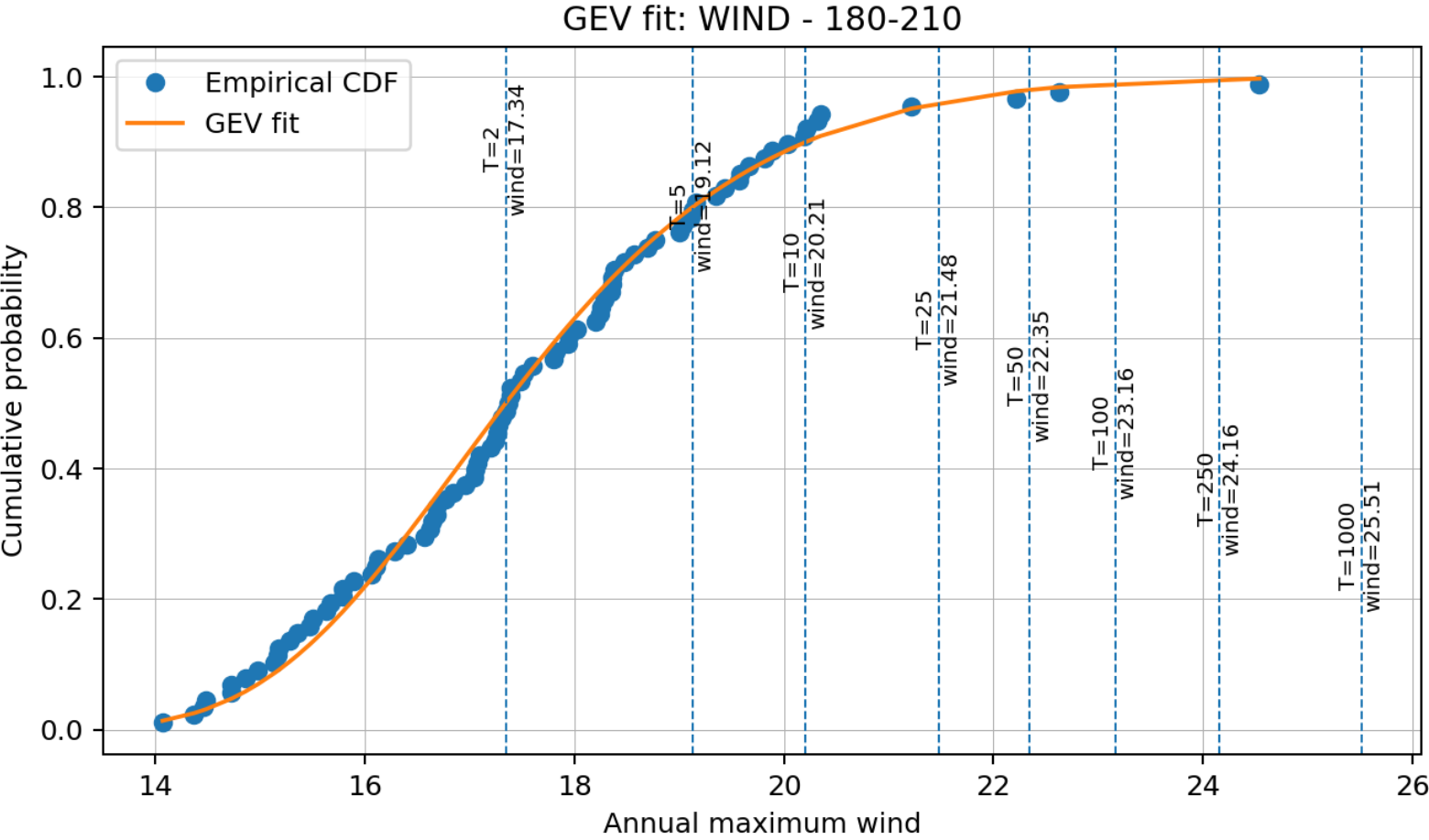


Wind sector 150-180

GEV fit: WIND - 150-180

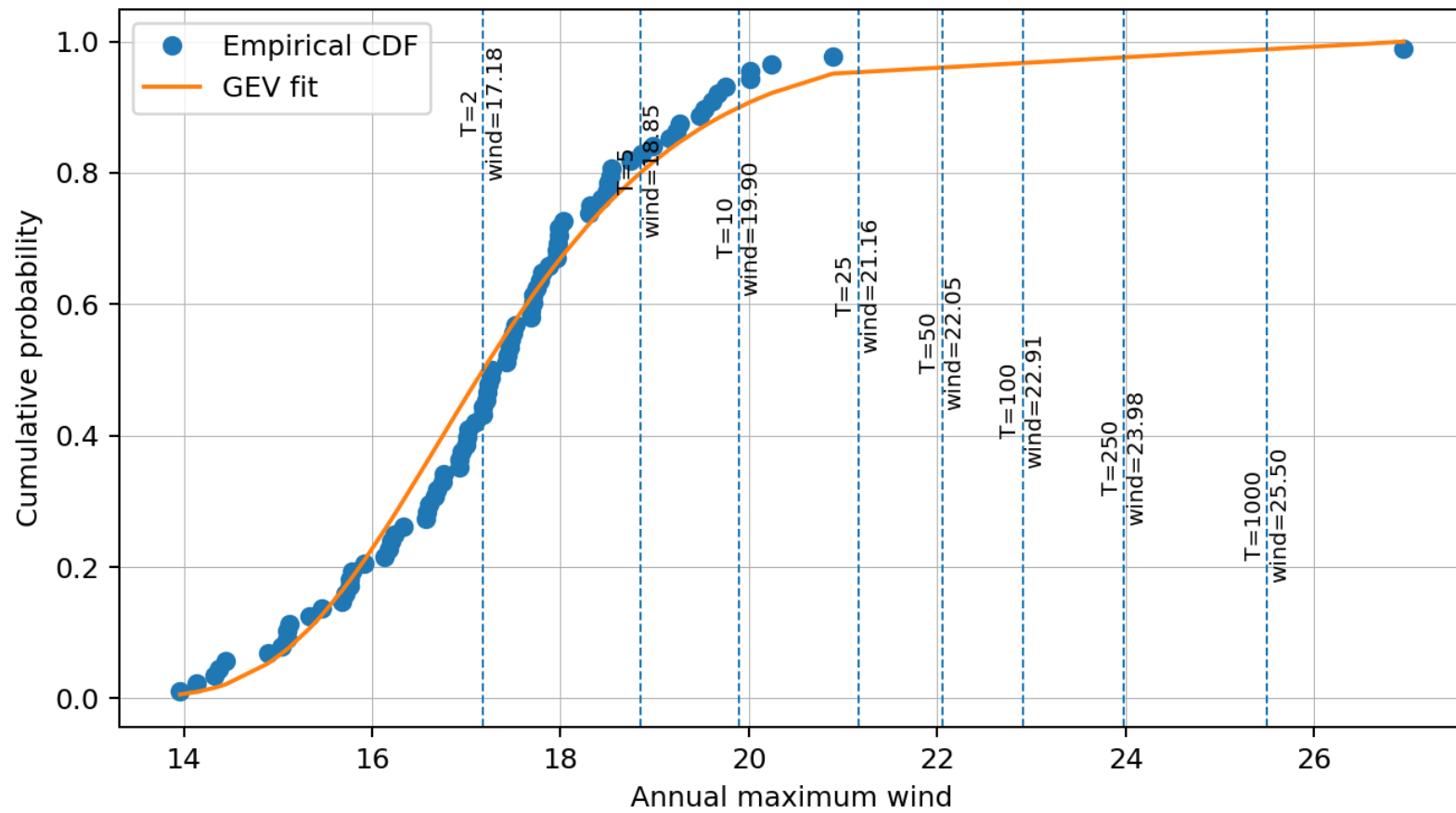


Wind sector 180-210

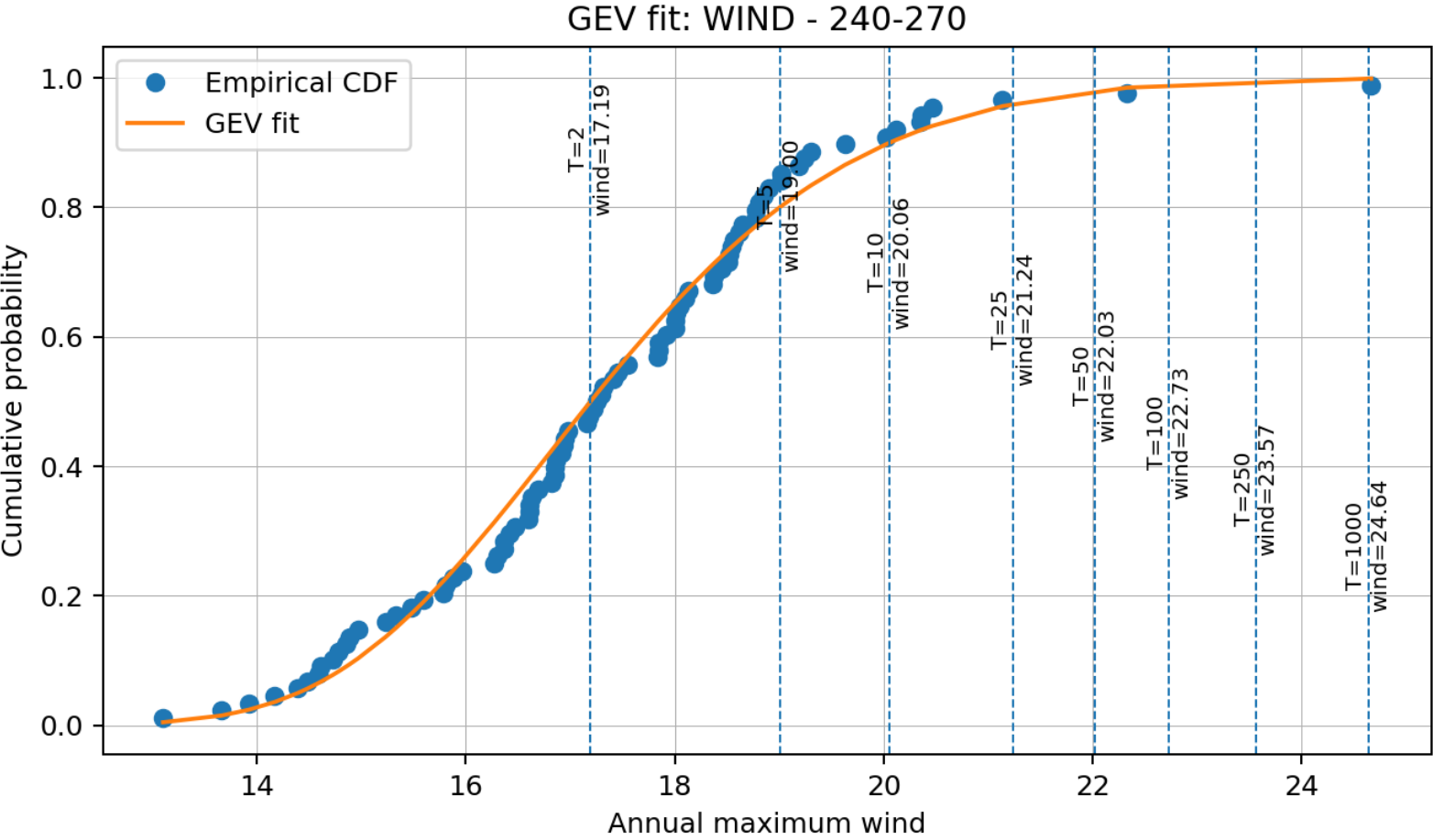


Wind sector 210-240

GEV fit: WIND - 210-240

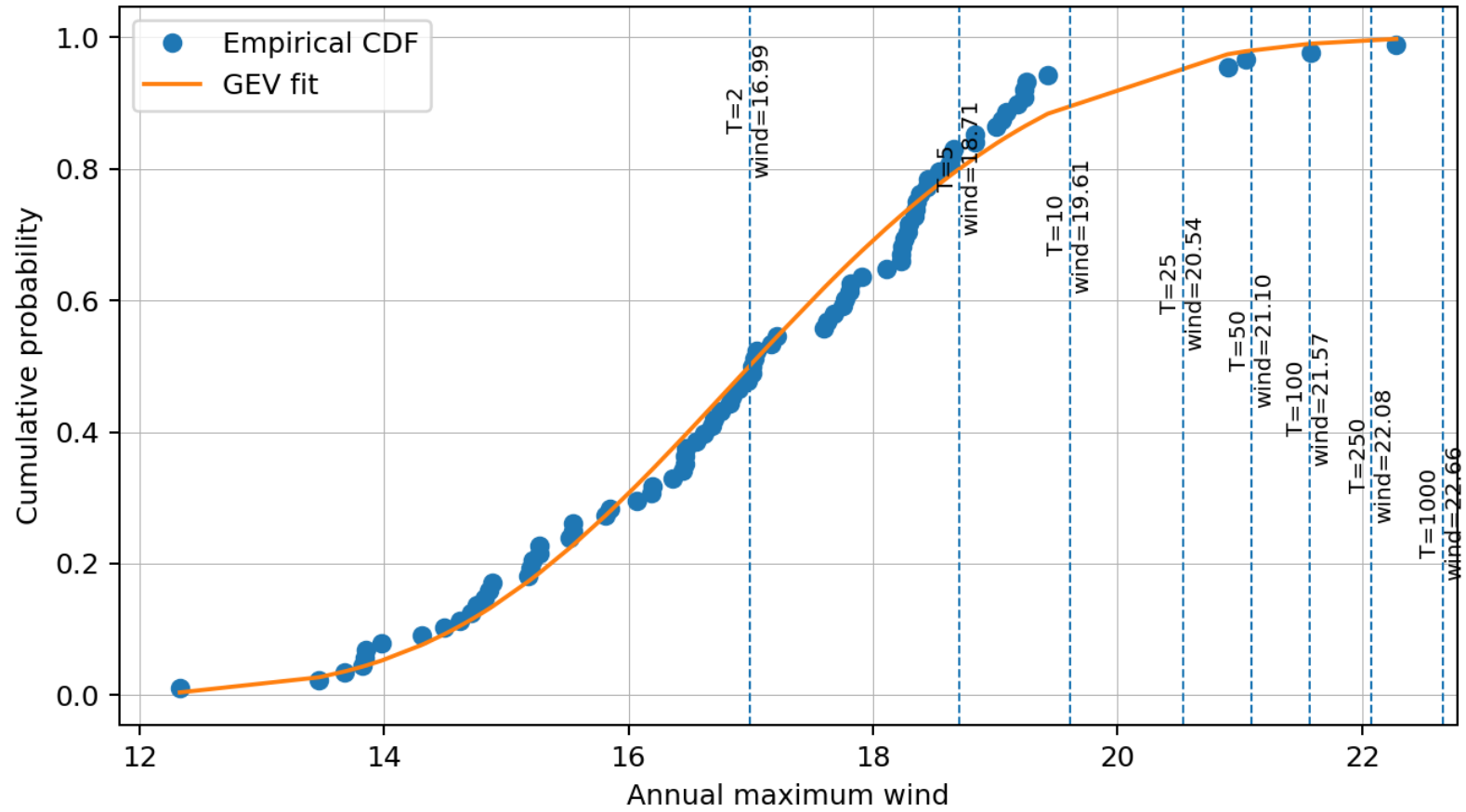


Wind sector 240-270

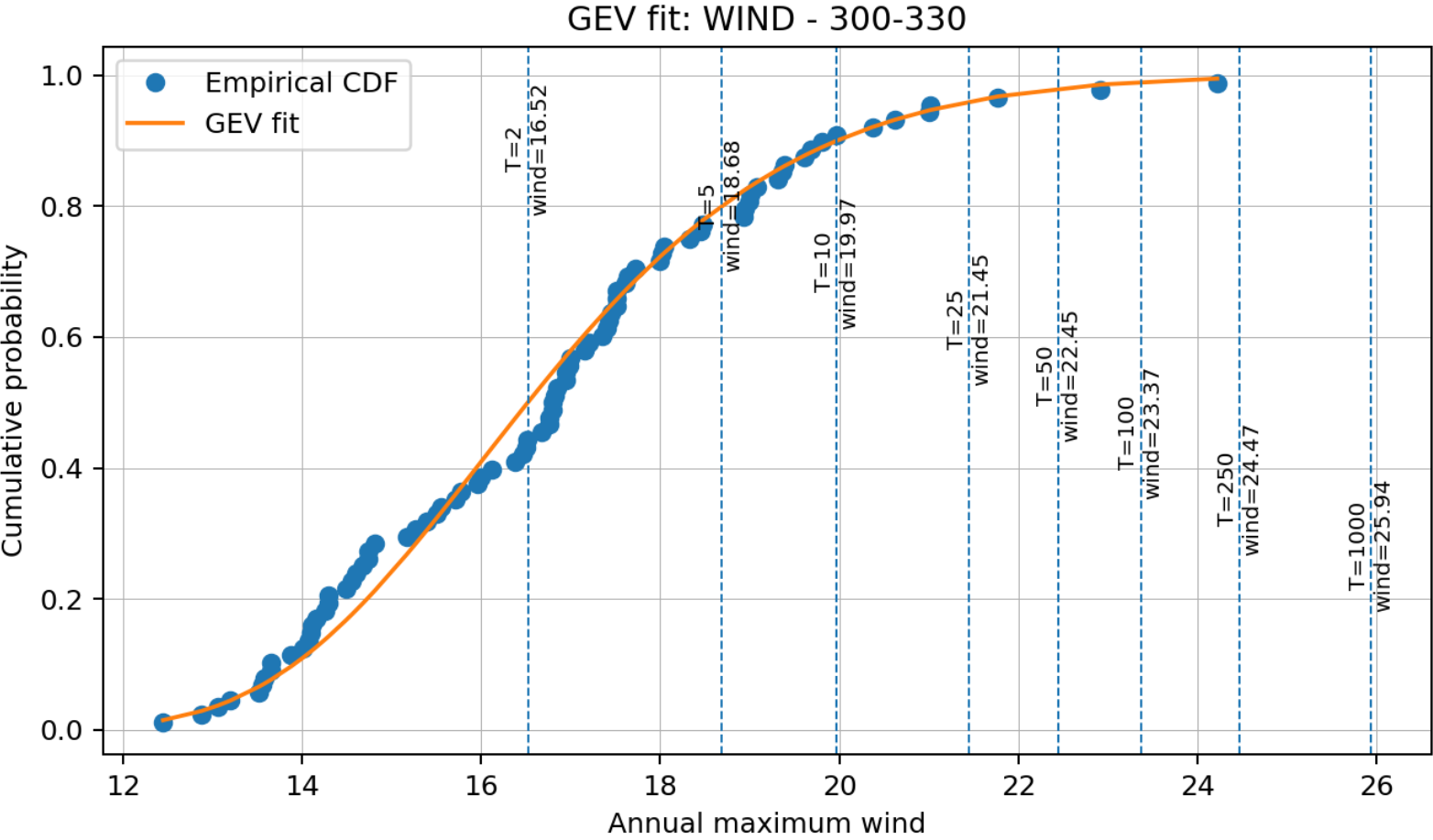


Wind sector 270-300

GEV fit: WIND - 270-300

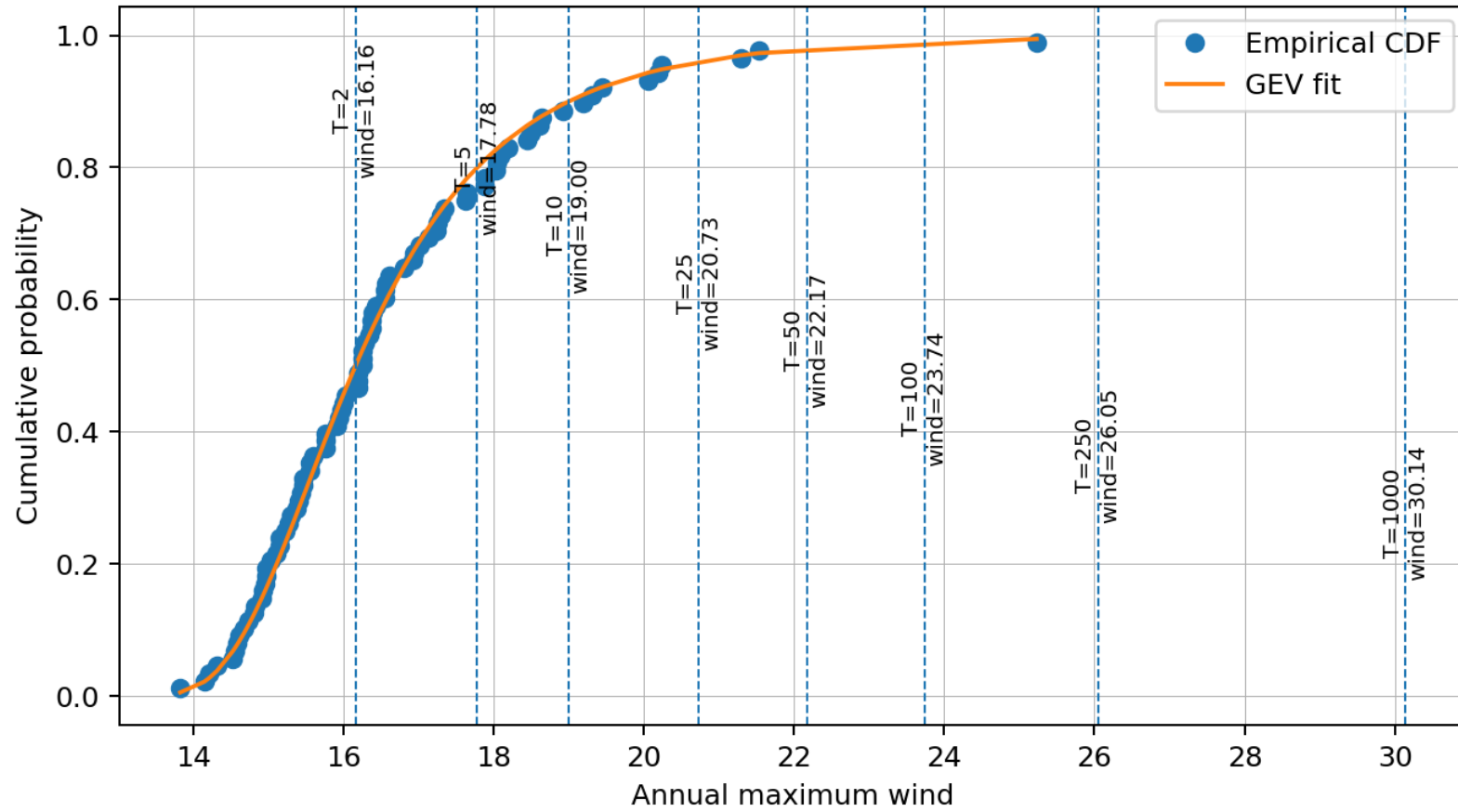


Wind sector 300-330



Wind sector 330-360

GEV fit: WIND - 330-360



6) Joint distribution: swl vs mwd (%)

Index	0-30	30-60	60-90	90-120	120-150	150-180	180-210	210-240	240-270	270-300	300-330	330-360	Total
0-1.5	0.332	0.059	0.021	0.016	0.022	0.027	0.102	0.378	1.416	6.096	11.957	6.644	27.071
1.5-3	0.814	0.075	0.039	0.031	0.055	0.163	0.522	1.396	3.425	11.553	19.559	16.008	53.640
3-4.5	0.140	0.001	0.001	0.002	0.004	0.062	0.230	0.644	1.648	4.560	5.588	2.579	15.458
4.5-6	0.001	0	0	0	0	0.011	0.025	0.112	0.418	1.155	1.210	0.222	3.154
6-7.5	0	0	0	0	0	0	0.001	0.018	0.063	0.233	0.245	0.014	0.574
7.5-9	0	0	0	0	0	0	0	0.001	0.012	0.039	0.038	0.001	0.092
9-10.5	0	0	0	0	0	0	0	0	0.003	0.004	0.003	0	0.010
10.5-12	0	0	0	0	0	0	0	0	0	0	0	0	0.001
Total	1.287	0.134	0.061	0.049	0.081	0.263	0.880	2.550	6.986	23.641	38.600	25.467	100

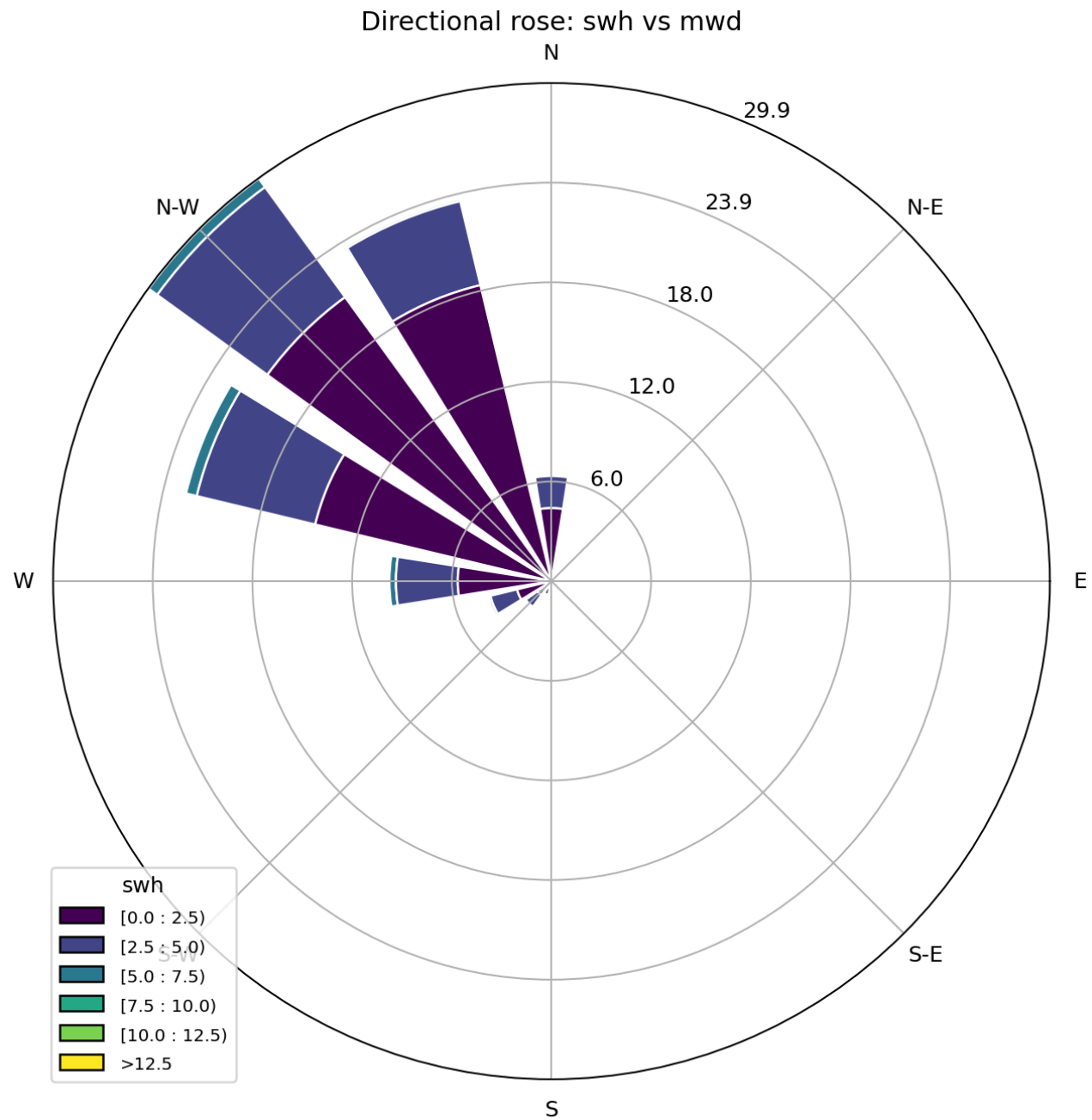
7) Joint distribution: swl vs mwp (%)

Index	3-4.5	4.5-6	6-7.5	7.5-9	9-10.5	10.5-12	12-13.5	13.5-15	15-16.5	16.5-18	Total
0-1.5	0.024	3.693	10.788	8.419	3.380	0.700	0.064	0.004	0	0	27.071
1.5-3	0	2.429	13.673	14.169	13.266	7.737	2.101	0.249	0.016	0	53.640
3-4.5	0	0	0.683	2.998	4.454	4.247	2.364	0.652	0.059	0.003	15.458
4.5-6	0	0	0	0.115	0.711	1.115	0.774	0.369	0.067	0.004	3.154
6-7.5	0	0	0	0	0.038	0.198	0.212	0.096	0.027	0.003	0.574
7.5-9	0	0	0	0	0.001	0.017	0.044	0.025	0.005	0	0.092
9-10.5	0	0	0	0	0	0.001	0.005	0.004	0	0	0.010
10.5-12	0	0	0	0	0	0.001	0	0	0	0	0.001
Total	0.024	6.122	25.144	25.701	21.849	14.014	5.565	1.398	0.174	0.010	100

8) Joint distribution: wind vs dwi (%)

Index	0-30	30-60	60-90	90-120	120-150	150-180	180-210	210-240	240-270	270-300	300-330	330-360	Total
0-3	1.254	1.015	0.841	0.757	0.746	0.808	0.971	1.051	1.151	1.279	1.491	1.551	12.917
3-6	4.282	2.012	1.478	1.292	1.042	1.530	2.174	2.134	1.936	2.039	3.450	6.471	29.841
6-9	5.464	1.090	0.845	0.789	0.577	1.445	2.380	2.021	1.701	1.491	2.097	9.377	29.276
9-12	5.342	0.366	0.182	0.223	0.303	1.138	1.701	1.438	1.097	0.891	0.979	6.604	20.265
12-15	2.163	0.061	0.011	0.013	0.120	0.544	0.630	0.542	0.421	0.337	0.338	1.576	6.756
15-18	0.121	0.001	0	0.001	0.037	0.131	0.134	0.111	0.089	0.078	0.076	0.081	0.860
18-21	0.001	0	0	0	0.003	0.010	0.015	0.010	0.011	0.009	0.014	0.007	0.079
21-24	0	0	0	0	0.001	0	0	0	0.001	0.001	0.001	0.001	0.004
24-27	0	0	0	0	0	0	0	0	0	0	0	0	0.001
27-30	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	18.628	4.547	3.357	3.076	2.828	5.605	8.006	7.307	6.405	6.125	8.447	25.668	100

9) Directional rose: swh vs mwd



10) Directional rose: wind vs dwi

